# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

### *Entropy And Spontaneity, Physical Significance Of Entropy*

**1.** For conversion C(graphite) → C(Diamond), the ΔS is :-
(A) Zero (B) Positive (C) Negative (D) Cannot be predicted
**CTS087**

**2.** For the process, $CO_2(s) \longrightarrow CO_2(g)$ :
(A) Both ΔH and ΔS are positive (B) ΔH is negative and ΔS is positive
(C) ΔH is positive and ΔS is negative (D) Both ΔH and ΔS are negative
**CTS089**

**3.** When disorder of a system increases, the change is said to be
(A) Exothermic (B) Non-spontaneous (C) Spontaneous (D) None of these
**CTS090**

**4.** Mixing of non-reacting gases is generally accompanied by
(A) Decrease in entropy (B) Increase in entropy
(C) Change in enthalpy (D) Change in free energy
**CTS091**

**5.** Which of the following processes are non spontaneous ?
(A) Burning of fossil fuel
(B) Decomposition of water into $H_2$ and $O_2$ gas at room temperature
(C) Spreading of perfume in a room
(D) Diffusion of gas from high pressure to low pressure
**CTS093**

**6.** How many of the following reaction entropy change is positive?
(A) $O_2(g) \rightarrow 2O(g)$ (B) $2O_3(g) \rightarrow 3O_2(g)$
(C) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ (D) $NaCl(s) \rightarrow Na^{\oplus}(aq) + Cl^{\ominus}(aq)$
(E) $C_2H_5OH(l) \rightarrow C_2H_5OH(g)$ (F) $Ag^{\oplus}(aq) + Cl^{\ominus}(aq) \rightarrow AgCl(s)$
(A) 4 (B) 5 (C) 6 (D) 2
**CTS098**

### *Second Law of Thermodynamics*

**7.** The second law of thermodynamics defines entropy(S), a state function ?
Which of the following statement is valid for entropy of an isolated system.
(A) For an isolated system at equilibrium, both energy and disorder are fixed.
(B) For an isolated system undergoing a spontaneous change, entropy decreases.
(C) For an isolated system undergoing a spontaneous change, both energy and entropy increases.
(D) For an isolated system undergoing spontaneous change, entropy can never decrease.
**CTS105**

### *Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process*

**8.** Which of the following is zero during adiabatic expansion of the gas
(A) ΔT (B) ΔS (C) ΔU (D) None of these
**CTS088**

**9.** Find the change in entropy (in cal/k) of 1 mole of $O_2$ gas ($C_V = 5/2\ R$), when it is heated from 300 K to 400 K isobarically ?
(A) 2.1 cal/K (B) 1.5 cal/K (C) 1.2 cal/K (D) 1.7 cal/K
**CTS095**

**10.** 5 mole of an ideal gas expands reversibly from a volume of 8 $dm^3$ to 80 $dm^3$ at a constant temperature of 27°C. The change in entropy is :–
(A) 41.57 $JK^{-1}$ (B) –95.73 $JK^{-1}$ (C) 95.73 $JK^{-1}$ (D) –41.57 $JK^{-1}$
**CTS096**



For More Join: @IITJEE_Advanced

**11.** Two blocks of equal masses and heat capacity but at different temperatures are taken in an adiabatic vessel, what is the change in entropy when they both attain thermal equilibrium?
(A) $\Delta S > 0$ (B) $\Delta S < 0$ (C) $\Delta S = 0$ (D) None of these

**CTS107**

**12.** One mole of an ideal diatomic gas ($C_V$ = 5 cal) was transformed from initial 25°C and 1L to the state when temperature is 100°C and volume 10L. The entropy change of the process can be expressed as (R = 2 calories /mol / k) :-
(A) $3\ell n \frac{298}{373} + 2\ell n\ 10$ (B) $5\ell n \frac{373}{298} + 2\ell n\ 10$
(C) $7\ell n \frac{373}{298} + 2\ell n \frac{1}{10}$ (D) $5\ell n \frac{373}{298} + 2\ell n \frac{1}{10}$

**CTS109**

**13.** Two moles of an ideal monoatomic gas is expanded adiabatically from 10L, 27°C to 80L. Calculate $\Delta S_{sys}$, $\Delta S_{surr}$ and $\Delta S_{univ.}$, if the process is performed irreversibly, against a constant external pressure of 0.1 atm.
(A) 35.9 J/K (B) 33.9 J/K (C) 32.9 J/K (D) 37.9 J/K

**CTS116**

### *Entropy Change For Phase Transformation*

**14.** One mole of NaCl(s) on melting at it's melting point absorbed 32.76 kJ of heat and its entropy is increased by 30 $JK^{-1}\ mol^{-1}$. What is the melting point of sodium chloride ?
(A) 819 °C (B) 1000 °C (C) 1000 K (D) 850 °C

**CTS097**

**15.** If 900 J/g of heat is exchanged at boiling point of water, then what is increase in entropy
(A) 43.4 J/mole (B) 87.2 J/mole (C) 900 J/mole (D) Zero

**CTS100**

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**16.** The entropy change, in the conversion of one mole of liquid water at 373 K to vapour at the same temperature is (Latent heat of vaporization of water, $\Delta H_{vap}$ = 2.257 kJ/g)
(A) 105.9 $JK^{-1}\ mol^{-1}$ (B) 107.9 $JK^{-1}\ mol^{-1}$ (C) 108.9 $JK^{-1}\ mol^{-1}$ (D) 109.9 $JK^{-1}\ mol^{-1}$

**CTS101**

### *Entropy Change For Chemical Reaction*

**17.** The entropy values (in $JK^{-1}\ mol^{-1}$) of $H_{2(g)}$ = 130.6, $Cl_{2(g)}$ = 223.0 and $HCl_{(g)}$ = 186.7 at 298 K and 1 atm pressure. Then entropy change for the reaction $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$ is
(A) + 540.3 (B) + 727.3 (C) – 166.9 (D) + 19.8

**CTS092**

### *Absolute Entropy And Third Law Of Thermodynamics, Residual Entropy, T-S Diagrams*

**18.** The area under a plot of $C_p/T$ v/s T at constant pressure represents -
(A) $\Delta S$ (B) $\Delta U$ (C) $\Delta H$ (D) q

**CTS110**

### *Gibbs Free Energy And Spontaneity, Calculation Of Change In G ( $\Delta$ G)*

**19.** A reaction has $\Delta H = -33$ kJ and $\Delta S = -58$ J/K. This reaction would be :-
(A) Spontaneous at all temperature
(B) Non-spontaneous at all temperature
(C) Spontaneous above a certain temperature only
(D) Spontaneous below a certain temperature only

**CTS117**

**20.** For a reaction at 25°C enthalpy change and entropy changes are $-11.7 \times 10^3$ J $mol^{-1}$ and $-105$ J $mol^{-1}\ K^{-1}$ respectively. What is the change in Gibbs free energy (kJ) :-
(A) 15.05 kJ (B) 19.59 kJ (C) 2.55 kJ (D) 22.55 kJ

**CTS118**



For More Join: @IITJEE_Advanced

**21.** $A + B \rightarrow C + D$
$\Delta H = -10{,}000\ J\ mol^{-1}$
$\Delta S = -33.3\ J\ mol^{-1}\ K^{-1}$
At what temperature the reaction will occur spontaneously from left to right?
(A) = 300.3 K (B) > 300.3 K (C) < 300.3 K (D) None of these
**CTS119**

**22.** Suppose that a reaction has $\Delta H = -40$ kJ and $\Delta S = -50$ J/K. At what temperature range will it change from spontaneous to non spontaneous?
(A) 0.8 K to 1 K (B) 799 K to 800 K (C) 800 K to 801 K (D) 799 K to 801 K
**CTS120**

**23.** $\Delta H$ and $\Delta S$ for the reaction
$Br_2(\ell) + Cl_2(g) \rightarrow 2BrCl(g)$ are 29.37 kJ and 104.0 $JK^{-1}$
respectively above what temperature will this reaction become spontaneous?
(A) 282.4°C (B) 282.4 K (C) 555.4 K (D) 555.4°C
**CTS125**

**24.** At 0°C, ice and water are in equilibrium and $\Delta H = 6kJ\ mol^{-1}$. For this process :
$H_2O(s) \rightleftharpoons H_2O(l)$
The values of $\Delta S$ and $\Delta G$ for conversion of ice into liquid water at 0°C are :-
(A) $-21.9\ J\ K^{-1}\ mol^{-1}$ and 0 (B) $0.219\ J\ K^{-1}\ mol^{-1}$ and 0
(C) $21.9\ J\ K^{-1}\ mol^{-1}$ and 0 (D) $0.0219\ J\ K^{-1}\ mol^{-1}$ and 0
**CTS126**

### *ΔG-Gibbs Free Energy Change (Non-PV Work) And Equilibrium*

**25.** $\Delta G°$ for the reaction $x + y \rightleftharpoons z$ is −4.606 kcal. The value of equilibrium constant of the reaction at 227°C is :-
(A) 100 (B) 10 (C) 2 (D) 0.01
**CTS122**

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**26.** Acetic acid $CH_3COOH$ can form a dimer $(CH_3COOH)_2$ in the gas phase
$2CH_3COOH(g) \rightleftharpoons (CH_3COOH)_2(g)$ ; $\Delta H_{298K} = -66.5$ kJ
if at 25°C, the equilibrium constant for this reaction is $10^3$ then calculate $\Delta S°$ for this reaction :-
(A) $-0.563\ kJ\ mol^{-1}\ K^{-1}$ (B) $-0.763\ kJ\ mol^{-1}\ K^{-1}$
(C) $+0.563\ kJ\ mol^{-1}\ K^{-1}$ (D) $-0.165\ kJ\ mol^{-1}\ K^{-1}$
**CTS124**

### *Carnot Cycle*

**27.** A heat engine absorbs 760 kJ heat from a source at 380K. It rejects 650 kJ of heat to sink at 280 K, it represents which cycle?
(A) Reversible cycle (B) Non-reversible cycle
(C) Impossible cycle (D) None of these
**CTS103**

**28.** A reversible heat engine A (based on Carnot cycle) absorbs heat from a reservoir at 1000 K and rejects heat to a reservoir at $T_2$. A second reversible engine B (based on Carnot cycle) absorbs the same amount of heat as rejected by the engine A, from the reservoir at $T_2$ and rejects energy to a reservoir at 360K.
If the efficiencies of engines A and B are the same then the temperature $T_2$ is :-
(A) 680 K (B) 640 K (C) 600 K (D) 670 K
**CTS104**

**29.** The Carnot's engine's efficiency depends on the absolute temperature of the_____
(A) Source (B) Sink
(C) Both source and sink (D) Neither source nor sink
**CTS106**



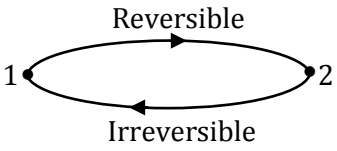
For More Join: @IITJEE_Advanced

# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

### *Second Law Of Thermodynamics*

**1.** For the cyclic process given below, which of the following relations are correct?




(A) $\Delta S = S_2 - S_1 = \int_1^2 \frac{\delta q_{rev}}{T}$

(B) $\Delta S = S_1 - S_2 = \int_2^1 \frac{\delta q_{irr}}{T}$

(C) $\Delta S_{cycle} = 0 = \int_1^2 \frac{\delta q_{rev}}{T} + \int_2^1 \frac{\delta q_{irr}}{T}$

(D) $\Delta S_{cycle} = 0 > \int_1^2 \frac{\delta q_{rev}}{T} + \int_2^1 \frac{\delta q_{irr}}{T}$

**CTS173**

### *Gibbs Free Energy And Spontaneity, Calculation Of Change In G ( $\Delta G$)*

**2.** Which of the following processes are expected to be spontaneous at low temperature and nonspontaneous at high temperature?

(A) Dissociation of $PCl_5(g)$ into $PCl_3(g)$ and $Cl_2(g)$.

(B) Formation of $H_2O(l)$ from $H_2(g)$ and $O_2(g)$.

(C) Formation of $H_2O(g)$ from $H_2(g)$ and $O_2(g)$.

(D) Freezing of water.

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**CTS174**

**3.** The normal boiling point of a liquid is 350 K and $\Delta H_{vap}$ is 35 kJ/mol. Assume that $\Delta H_{vap}$ is independent from temperature and pressure. The correct statement(s) is/are

(A) $\Delta S_{vap} > 100$ J/K-mol at 350 K and 0.5 atm.

(B) $\Delta G_{vap} < 0$ at 350 K and 0.5 atm.

(C) $\Delta S_{vap} > 100$ J/K-mol at 350 K and 2.0 atm.

(D) $\Delta G_{vap} > 0$ at 350 K and 2.0 atm.

**CTS175**

## MATRIX MATCH TYPE QUESTION

### *Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process*

**4.** Match the columns

| Column -I | | Column -II | |
|---|---|---|---|
| (A) | Reversible isothermal expansion of an ideal gas | (P) | $\Delta S_{sys} = 0$<br>$\Delta S_{surr} = 0$<br>$\Delta S_{total} = 0$ |
| (B) | Reversible adiabatic expansion of an ideal gas | (Q) | $\Delta H = \Delta U = \Delta S_{total} = 0$ |
| (C) | Adiabatic free expansion | (R) | $\Delta S_{total} > 0$ |
| | | (S) | $q = 0$ |

**CTS176**



For More Join: @IITJEE_Advanced

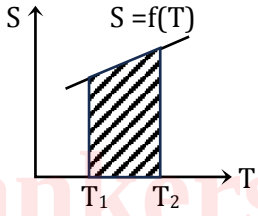
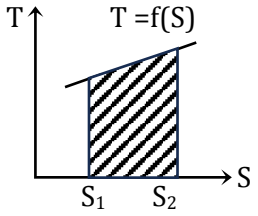
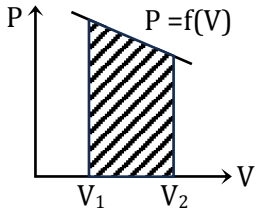
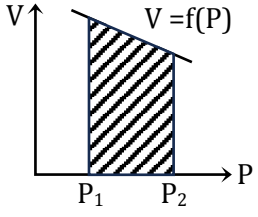
**5.** Match the columns.

| | | Column -II | |
|---|---|---|---|
| (A) | For the process: $A(l) \rightarrow A(s)$, $\Delta H$ and $\Delta V$ may be | (P) | −ve, +ve |
| (B) | $A_2(s) + B_2(g) \rightarrow C_2(s) + D_2(s)$, $\Delta H$ and $\Delta G$ may be | (Q) | +ve, −ve |
| (C) | $A_2(g) \rightarrow B(g) + C(g)$, $E_{a(forward)} = 50$ kJ/mol, $E_{a(backward)} = 40$ kJ/mol $\Delta H$ and $\Delta G$ at very high temperature are | (R) | +ve, +ve |
| (D) | $A(g) \rightarrow B(g)$ $\Delta H$ and $\Delta G$ at very low temperature are | (S) | −ve, −ve |

**CTS177**

## *Absolute Entropy And Third Law of Thermodynamics, Residual Entropy, T-S Diagrams*

**6.** Match Column I (graph) with the physical quantity in Column II (area shown in the graph represents the magnitude of).

| Column -I | | Column -II | |
|---|---|---|---|
| (A) |   | (P) | q |
| (B) |   | (Q) | w |
| (C) |   | (R) | $(\Delta G)_T$ |
| (D) |   | (S) | $(\Delta G)_P$ |

**CTS178**

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## ASSERTION & REASON TYPE QUESTIONS

### *Entropy And Spontaneity, Physical Significance of Entropy*

Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason).
Examine the statements carefully and mark the correct answer according to the instructions given below :

(A) If both the statements are TRUE and STATEMENT-2 is the correct explanation of STATEMENT-1
(B) If both the statements are TRUE but STATEMENT-2 is NOT the correct explanation of STATEMENT-1
(C) If STATEMENT-1 is TRUE and STATEMENT-2 is FALSE
(D) If STATEMENT-1 is FALSE and STATEMENT-2 is TRUE

**7.** **STATEMENT-1 :** Many endothermic reactions that are not spontaneous at room temperatures become spontaneous at high temperature.
**STATEMENT-2 :** $\Delta H^{\circ}$ of the endothermic reaction may increases with increase in temperature.

(A) If both the statements are TRUE and STATEMENT-2 is the correct explanation of STATEMENT-1
(B) If both the statements are TRUE but STATEMENT-2 is NOT the correct explanation of STATEMENT-1
(C) If STATEMENT-1 is TRUE and STATEMENT-2 is FALSE
(D) If STATEMENT-1 is FALSE and STATEMENT-2 is TRUE

**CTS179**

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For More Join: @IITJEE_Advanced

# EXERCISE (S)

## Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

### *Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process*

**1.** One mole of an ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate $\Delta S_{total}$ when the process is carried out reversibly.

**CTS029**

### *Entropy Change For Phase Transformation*

**2.** At 0°C, water and ice are at equilibrium at 1 atm pressure. The value of $\Delta H_{fusion}$ for ice is (in kJ/mole).
(Given $\Delta S_{freezing}$ of water at 1 atm & 273 K is –20 JK mol)
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

**CTS024**

### *Gibbs Free Energy And Spontaneity, Calculation Of Change In G (ΔG)*

**3.** Find magnitude of difference in standard Gibb's energy of formation of $H_2O(g)$ & $H_2O(l)$ at 300K in kJ/mol if Vapour pressure of $H_2O$ at 300 K is 30 torr. [Use : 1 bar = 75 cm Hg]
[Give your answer as nearest integer]

**CTS021**

**4.** The standard entropies of $H_2(g)$ and H(g) are 130 and 115 J $mol^{-1}K^{-1}$ respectively at 300K. Using the data given below calculate the bond energy of $H_2$(in kJ/mol) :
$H_2(g) \longrightarrow 2H(g)$ ; $\Delta G° = 406$ kJ/mol

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**CTS028**

### *ΔG-Gibbs Free Energy Change (Non-PV Work) And Equilibrium*

**5.** Identify the number of processes with $\Delta G = 0$ :
(a) $H_2O(l, 0°C, 1atm) \rightarrow H_2O(g, 0°C, 1 atm)$
(b) $H_2O(l, 100°C, 1 atm) \rightarrow H_2O(g, 100°C, 1 atm)$
(c) $H_2O(l, 27°C, 0.04 atm) \rightarrow H_2O(g, 27°C, 0.04 atm)$
Vapour pressure of water at 27°C = 0.04 atm
(d) $H_2O(s, –10°C, 0.01 atm) \rightarrow H_2O(g, –10°C, 0.01 atm)$
Vapour pressure of ice at –10°C = 0.01 atm
(e) $H_2O(l, 0°C, 1atm) \rightarrow H_2O(s, 0°C, 1 atm)$
(f) $H_2O(l, 100°C, 1atm) \rightarrow H_2O(g, 100°C, 0.1 atm)$
(g) $H_2O(l, 0°C, 1atm) \rightarrow H_2O(s, 0°C, 5 atm)$
270° C = 0.04 atm

**CTS023**

## Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.



For More Join: @IITJEE_Advanced

### *Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process*

**6.** A diatomic ideal gas is expanded isothermally to 32 times of it's initial volume then it is cooled to restore to initial entropy at constant volume. Calculate ratio of initial temperature to final temperature

**CTS127**

**7.** Find the change in entropy (in cal/K) of 1 mole of $O_2$ gas ($C_V$ = 5/2 R), when it is
(a) heated from 300 K to 400 K isobarically
(b) heated from 300 K to 400 K isochorically (Given : ln 3 = 1.1, ln2 = 0.7)

**CTS129**

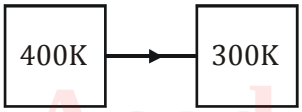
**8.** The entropy change (in J/K) involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 $dm^3$ to a volume of 100 $dm^3$ at 27°C is :-

**CTS130**

**9.** 1 mole of ideal monoatomic gas is heated by supplying 5 kJ heat from a reservoir maintained at 400 K from 300 K to 400 K. In the process volume of gas increased from 1 L to 10 L. Find $\Delta S_{total}$ (in J/K -mol) in the process
Use : $\ln\left(\frac{4}{3}\right)$ = 0.3, *l*n 10 = 2.3, R = 8.3 J/K-mol and *l*n x = 2.3 log x

**CTS131**

**10.** 100 kJ heat is transferred from a large heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat and combinedly source and sink form isolated system.




Source Sink

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Find (a)$\Delta S_{source}$, (b) $\Delta S_{sink}$ and (c) $\Delta S_{total}$. Also comment on spontaneity of process.

**CTS128**

### *Gibbs Free Energy And Spontaneity, Calculation of Change In G (ΔG)*

**11.** Calculate the $\Delta G^{\circ}$ (in kJ) change at 300 K for the reaction;
$Br_2(\ell) + Cl_2(g) \longrightarrow 2BrCl(g)$. For the reaction $\Delta H^{\circ}$ = 29.3 kJ & the standard entropies of $Br_2(\ell)$, $Cl_2(g)$ & BrCl(g) at the 300 K are 150, 220, 240 J $mol^{-1}K^{-1}$ respectively.

**CTS132**

**12.** 5 mole $H_2O(\ell)$ at 373K and 1 atm is converted into $H_2O(g)$ at 373K and 5 atm. Calculate $\Delta G$ (in Kcal) for this process. [Given : R = 2Cal/K-mol, ln 5 = 1.6]

**CTS133**

**13.** For the reaction ;
$X(g) \rightarrow 3Y(g)$ ; $\Delta G$ = – 2.7 kcal , $\Delta S$ = 20 cal $K^{-1}$ at 300 K
Hence $\Delta U$ in kcal is_____ .

**CTS138**

### *ΔG-Gibbs Free Energy Change (Non-PV Work) And Equilibrium*

**14.** Standard entropy of $X_2$, $Y_2$ and $XY_3$ are 60, 40 and 50 $JK^{-1}mol^{-1}$, respectively. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H$ = –30 kJ. For the given reaction to be in equilibrium absolute temperature will be ?

**CTS134**



For More Join: @IITJEE_Advanced

**15.** For the reaction,

$SO_2(g) + 1/2\ O_2(g) \rightleftharpoons SO_3(g)$ ; $\Delta H°_{300} = -95$ kJ/mole,

$\Delta S°_{300} = -95.0$ J/K mole. Find the value of $(\ln K_p)$ for this reaction at 300 K.

Give your answer as nearest integer.

**CTS135**

**16.** For the reaction at 298 K

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$ ; $\Delta H° = -29.8$kcal; $\Delta S° = -0.1$ kcal/K

Calculate $\Delta G°$ (kcal/mole) and $K_p$ for the reaction ?

**CTS136**

**17.** For a dimerization reaction,

$2\ A(g) \rightarrow A_2(g)$ at 300 K, $\Delta U° = -20$kJ $mol^{-1}$, $\Delta S° = -30$ $JK^{-1}$ $mol^{-1}$

Calculate the value of $\Delta G°$ (in kJ/mol). (R = 8.3 J/mole-K)

**CTS137**

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# EXERCISE - JEE (Main) PYQ

**1.** The process with negative entropy change is: **[JEE (Main) 2019]**
(1) Dissolution of iodine in water
(2) Synthesis of ammonia from $N_2$ and $H_2$
(3) Dissolution of $CaSO_4(s)$ to $CaO(s)$ and $SO_3(g)$
(4) Sublimation of dry ice

**CTS045**

**2.** Two blocks of the same metal having same mass and at temperature $T_1$ and $T_2$, respectively. are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, $\Delta S$, for this process is : **[JEE (Main) 2019]**

(1) $2C_P \ln\left(\frac{T_1+T_2}{4T_1T_2}\right)$ (2) $2C_P \ln\left[\frac{(T_1+T_2)^{\frac{1}{2}}}{T_1T_2}\right]$ (3) $C_P \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$ (4) $2C_P \ln\left[\frac{T_1+T_2}{2T_1T_2}\right]$

**CTS038**

**3.** Ice at –5°C is heated to become vapor with temperature of 110°C at atmospheric pressure. The entropy change associated with this process can be obtained from : **[JEE (Main) 2025]**

(1) $\int_{268K}^{383K} C_p\,dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

(2) $\int_{268K}^{273K} \frac{C_{p,m}}{T}dT + \frac{\Delta H_m, fusion}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} \frac{C_{p,m}\,dT}{T} + \int_{373K}^{383K} \frac{C_{p,m}dT}{T}$

(3) $\int_{268K}^{383K} C_p\,dT + \frac{q_{rev}}{T}$

(4) $\int_{268K}^{273K} C_{p,m}dT + \frac{\Delta H_m, fusion}{T_f} + \frac{\Delta H_{m,vaporisation}}{T_b} + \int_{273K}^{373K} C_{p,m}dT + \int_{373K}^{383K} C_{p,m}dT$

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**CTS201**

**4.** Standard entropies of $X_2$, $Y_2$ and $XY_5$ are 70, 50 and 110 J $K^{-1}$ $mol^{-1}$ respectively. The temperature in Kelvin at which the reaction

$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightarrow XY_5$ ; $\Delta H^- = -35$ kJ $mol^{-1}$

Will be at equilibrium is ___ (Nearest integer) **[JEE (Main) 2025]**

**CTS202**

**5.** The true statement amongst the following is:
(1) Both $\Delta S$ and S are functions of temperature.
(2) S is not a function of temperature but $\Delta S$ is a function of temperature.
(3) Both S and $\Delta S$ are not functions of temperature.
(4) S is a function of temperature but $\Delta S$ is not a function of temperature.

**[JEE (Main) 2020]**

**CTS047**

**6.** The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapours at 383 K is : (Specific heat of water liquid and water vapour are 4.2 kJ $K^{-1}$ $kg^{-1}$ and 2.0 kJ $K^{-1}$ $kg^{-1}$ ; heat of liquid fusion and vapourisation of water are 344 kJ $kg^{-1}$ and 2491 kJ $kg^{-1}$, respectively).
(log 273 = 2.436, log 373 = 2.572, log 383 = 2.583) **[JEE (Main) 2019]**
(1) 7. 90 kJ $kg^{-1}$ $K^{-1}$ (2) 2.64 kJ $kg^{-1}$ $K^{-1}$ (3) 8.49 kJ $kg^{-1}$ $K^{-1}$ (4) 9.26 kJ $kg^{-1}$ $K^{-1}$

**CTS044**



For More Join: @IITJEE_Advanced

**7.** The standard enthalpy and standard entropy of decomposition of $N_2O_4$ to $NO_2$ are 55.0 kJ $mol^{-1}$ and 175.0 J/K/mol respectively. The standard free energy change for this reaction at 25°C in J $mol^{-1}$ is _____ (Nearest integer) **[JEE (Main) 2025]**

**CTS203**

**8.** The effect of temperature on spontaneity of reactions are represented as:

| | **ΔH** | **ΔS** | **Temperature** | **Spontaneity** |
|---|---|---|---|---|
| (A) | + | – | any T | Non spontaneous |
| (B) | + | + | low T | spontaneous |
| (C) | – | – | low T | Non spontaneous |
| (D) | – | + | any T | spontaneous |

**[JEE (Main) 2025]**

(1) (B) and (D) only (2) (A) and (D) only (3) (B) and (C) only (4) (A) and (C) only

**CTS204**

**9.** Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option. **[JEE (Main) 2025]**

(1) Both ΔH and ΔS are (+ve) (2) ΔH is (–ve) but ΔS is (+ve)
(3) ΔH is (+ve) but ΔS is (–ve) (4) Both ΔH and ΔS are (–ve)

**CTS205**

**10.** For the reaction at 298 K, $2A + B \rightarrow C$. $\Delta H = 400$ kJ $mol^{-1}$ and $\Delta S = 0.2$ kJ $mol^{-1}$ $K^{-1}$. The reaction will become spontaneous above_____ K. **[JEE (Main) 2024]**

**CTS180**

**11.** When $\Delta H_{vap}$ = 30 kJ/mol and $\Delta S_{vap}$=75 J $mol^{-1}$ $K^{-1}$, then the temperature of vapour, at one atmosphere is _____K. **[JEE (Main) 2024]**

**CTS181**

**12.** Which of the following is **not** correct? **[JEE (Main) 2024]**

(1) $\Delta G$ is negative for a spontaneous reaction (2) $\Delta G$ is positive for a spontaneous reaction
(3) $\Delta G$ is zero for a reversible reaction (4) $\Delta G$ is positive for a non-spontaneous reaction

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**CTS182**

**13.** Which of the following relations are correct?

(A) $\Delta U = q + p\Delta V$ (B) $\Delta G = \Delta H - T\Delta S$ (C) $\Delta S = \frac{q_{rev}}{T}$ (D) $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below : **[JEE (Main) 2023]**

(1) C and D only (2) B and C only (3) A and B only (4) B and D only

**CTS183**

**14.** The standard entropy change for the reaction
$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is –550 $JK^{-1}$ at 298 K.
[Given : The standard enthalpy change for the reaction is –165 kJ $mol^{-1}$]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer) **[JEE (Main) 2022]**

**CTS139**

**15.** For a given chemical reaction $A \rightarrow B$ at 300 K the free energy change is -49.4 kJ $mol^{-1}$ and the enthalpy of reaction is 51.4 kJ $mol^{-1}$. The entropy change of the reaction is _____ J $K^{-1}$ $mol^{-1}$. **[JEE (Main) 2021]**

**CTS049**

**16.** For the reaction ;
$A(l) \rightarrow 2B(g)$
$\Delta U$ = 2.1 kcal, $\Delta S$ = 20 cal $K^{-1}$ at 300 K
Hence $\Delta G$ in kcal is_____ . **[JEE (Main) 2020]**

**CTS046**



For More Join: @IITJEE_Advanced

**17.** Match **List-I** with **List-II**.

| | **List-I (Partial Derivatives)** | | **List-II (Thermodynamic Quantity)** |
|---|---|---|---|
| (A) | $\left(\frac{\partial G}{\partial T}\right)_P$ | (I) | $C_P$ |
| (B) | $\left(\frac{\partial H}{\partial T}\right)_P$ | (II) | $-S$ |
| (C) | $\left(\frac{\partial G}{\partial P}\right)_T$ | (III) | $C_V$ |
| (D) | $\left(\frac{\partial U}{\partial T}\right)_V$ | (IV) | V |

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (A)-(II), (B)-(I), (C)-(III), (D)-(IV) (2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(3) (A)-(I), (B)-(II), (C)-(IV), (D)-(III) (4) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

**CTS206**

**18.** For a certain reaction at 300K, K = 10, then $\Delta G^\circ$ for the same reaction is $-$_____$\times 10^{-1}$ kJ $mol^{-1}$. (Given R = 8.314 $JK^{-1}$ $mol^{-1}$) **[JEE (Main) 2024]**

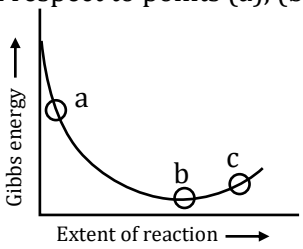
**CTS184**

**19.** For complete combustion of ethene.

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$ the amount of heat produced as measured in bomb calorimeter is 1406 kJ $mol^{-1}$ at 300K. The minimum value of $T\Delta S$ needed to reach equilibrium is $(-)$ _____kJ. (Nearest integer) Given : R = 8.3 $JK^{-1}$ $mol^{-1}$ **[JEE (Main) 2023]**

**CTS141**

**20.** Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is.....




A. Reaction is spontaneous at (a) and (b)

B. Reaction is at equilibrium at point (b) and non-spontaneous at point (c)

C. Reaction is spontaneous at (a) and non-spontaneous at (c)

D. Reaction is non-spontaneous at (a) and (b) **[JEE (Main) 2023]**

**CTS185**

**21.** For the chemical reaction $X \rightleftharpoons Y$, the standard reaction Gibbs energy depends on temperature T (in K) as : $\Delta_r G^\circ$ (in kJ $mol^{-1}$) = $120 - \frac{3}{8}T$

The major component of the reaction mixture at T is : **[JEE (Main) 2019]**

(1) X if T = 315 K (2) X if T = 350 K (3) Y if T = 300 K (4) Y if T = 280 K

**CTS039**

**22.** A process has $\Delta H$ = 200 $Jmol^{-1}$ and $\Delta S$ = 40 $JK^{-1}mol^{-1}$. Out of the values given below, choose the minimum temperature above which the process will be spontaneous : **[JEE (Main) 2019]**

(1) 5 K (2) 4 K (3) 20 K (4) 12 K

**CTS043**



For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Advanced) PYQ

**1.** For the process

$H_2O(l) \rightarrow H_2O(g)$

at T = 100°C and 1 atmosphere pressure, the correct choice is **[JEE (Advanced) 2014]**

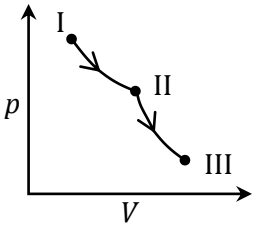
(A) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} > 0$ (B) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} < 0$

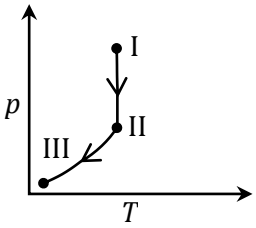
(C) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} > 0$ (D) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} < 0$

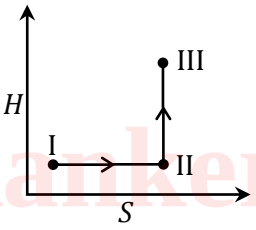
**CTS071**

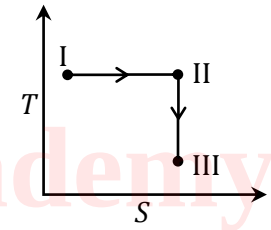
**2.** An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state II to state III. The correct plot(s) representing the changes from state I to state III is(are)

(p : pressure, V : volume, T : temperature, H : enthalpy, S : entropy) **[JEE (Advanced) 2021]**

(A) 


(B) 


(C) 


(D) 


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**CTS085**

**3.** One mole of an ideal gas at 300 K in thermal contact with surroundings expands isothermally from 1.0 L to 2.0 L against a constant pressure of 3.0 atm. In this process, the change in entropy of surroundings ($\Delta S_{surr}$) in J $K^{-1}$ is -

(1 L atm = 101.3 J) **[JEE (Advanced) 2016]**

(A) 5.763 (B) 1.013 (C) –1.013 (D) –5.763

**CTS073**

**4.** Match the transformations in **Column-I** with appropriate option in **Column-II**

| **Column-I** | **Column-II** |
|---|---|
| (1) $CO_2(s) \rightarrow CO_2(g)$ | (p) phase transition |
| (2) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ | (q) allotropic change |
| (3) $2\dot{H} \rightarrow H_2(g)$ | (r) $\Delta H$ is positive |
| (4) $P_{(white, solid)} \rightarrow P_{(red,solid)}$ | (s) $\Delta S$ is positive |
| | (t) $\Delta S$ is negative |

(A) (A) → (p, r, s) ; (B) → (r, s) ; (C) → (t) ; (D) → (p, q, t)

(B) (A) → (p, r) ; (B) → (s) ; (C) → (t) ; (D) → (q, t)

(C) (A) → (r, s) ; (B) → (r) ; (C) → (s) ; (D) → (p, q)

(D) (A) → (p, s) ; (B) → (q, s) ; (C) → (p) ; (D) → (p, t)

**[JEE (Advanced) 2011]**

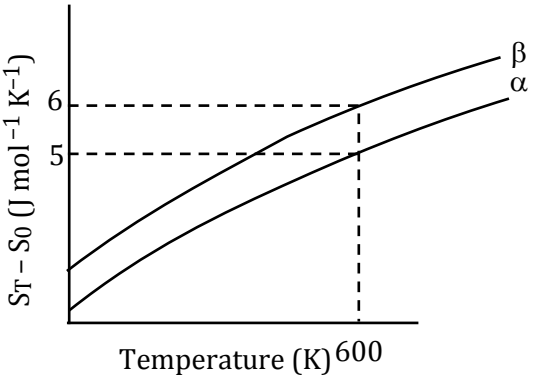
**CTS069**



For More Join: @IITJEE_Advanced

**"PARAGRAPH I" (Q.5 & 6)**

The entropy versus temperature plot for phases α and β at 1 bar pressure is given.
$S_T$ and $S_0$ are entropies of the phases at temperatures T and 0 K, respectively.




The transition temperature for α to β phase change is 600 K and $C_{P,\beta} - C_{P,\alpha} = 1$ J mol$^{-1}$ K$^{-1}$. Assume $(C_{P,\beta} - C_{P,\alpha})$ is independent of temperature in the range of 200 to 700 K. $C_{P,\alpha}$ and $C_{P,\beta}$ are heat capacities of α and β phases, respectively.

**[JEE (Advanced) 2023]**

**5.** The value of entropy change, $S_\beta - S_\alpha$ (in J mol$^{-1}$ K$^{-1}$), at 300 K is __.
[Use : ln 2 = 0.69
Given : $S_\beta - S_\alpha = 0$ at 0 K]

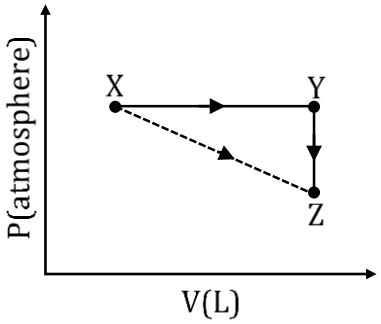
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**CTS144**

**6.** The value of enthalpy change, $H_\beta - H_\alpha$ (in J mol$^{-1}$), at 300 K is __.

**CTS145**

**7.** For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and was work done]




**[JEE (Advanced) 2012]**

(A) $\Delta S_{x\to z} = \Delta S_{x\to y} + \Delta S_{y\to z}$

(B) $W_{x\to z} = W_{x\to y} + W_{y\to z}$

(C) $W_{x\to y\to z} = W_{x\to y}$

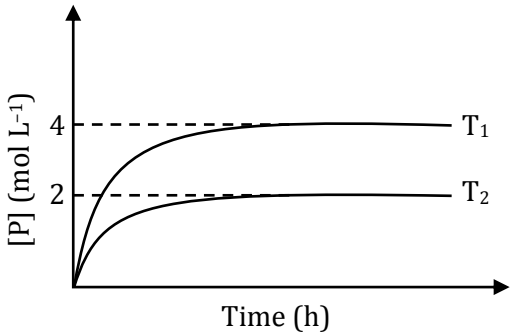
(D) $\Delta S_{x\to y\to z} = \Delta S_{x\to y}$

**CTS070**



For More Join: @IITJEE_Advanced

**8.** In a one-litre flask, 6 moles of A undergoes the reaction A (g) $\rightleftharpoons$ P (g). The progress of product formation at two temperatures (in Kelvin), $T_1$ and $T_2$, is shown in the figure:



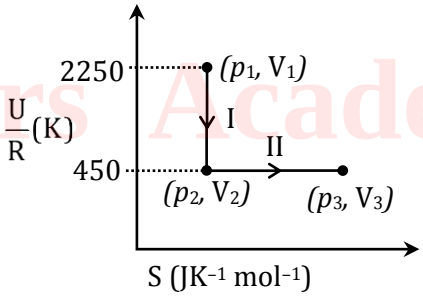

If $T_1 = 2T_2$ and $(\Delta G_2^{\ominus} - \Delta G_1^{\ominus}) = RT_2 \ln x$, then the value of x is _____.

[$\Delta G_1^{\ominus}$ and $\Delta G_2^{\ominus}$ are standard Gibb's free energy change for the reaction at temperatures $T_1$ and $T_2$, respectively.]

**[JEE (Advanced) 2023]**

**CTS143**

**9.** One mole of an ideal gas at 900 K, undergoes two reversible processes, I followed by II, as shown below. If the work done by the gas in the two processes are same, the value of $\ln \frac{V_3}{V_2}$ is _____.




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(U: internal energy, S: entropy, p: pressure, V: volume, R: gas constant)

(Given: molar heat capacity at constant volume, $C_{v,m}$ of the gas is $\frac{5}{2}$ R)

**[JEE (Advanced) 2021]**

**CTS086**

**10.** Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

At 298 K : $\Delta_f H°(SnO_2(s)) = -581.0$ kJ mol$^{-1}$, $\Delta_f H°(CO_2(g)) = -394.0$ kJ mol$^{-1}$

$S°(SnO_2(s)) = 56.0$ J K$^{-1}$ mol$^{-1}$, $S°(Sn(s)) = 52.0$ J K$^{-1}$ mol$^{-1}$,

$S°(C(s)) = 6.0$ J K$^{-1}$ mol$^{-1}$ , $S°(CO_2(g)) = 210.0$ J K$^{-1}$ mol$^{-1}$.

Assume that the enthalpies and the entropies are temperature independent.

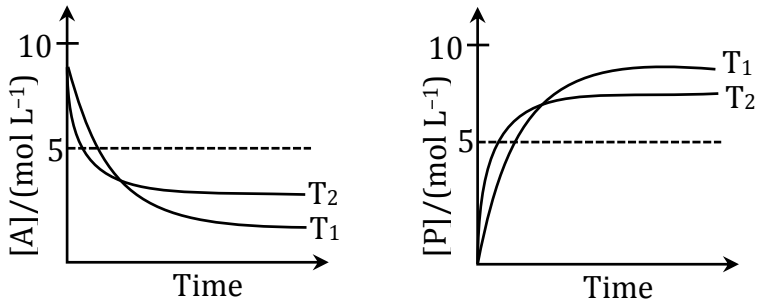
**[JEE (Advanced) 2020]**

**CTS082**



For More Join: @IITJEE_Advanced

**11.** For a reaction, $A \rightleftharpoons P$, the plots of [A] and [P] with time at temperatures $T_1$ and $T_2$ are given below.




If $T_2 > T_1$, the correct statement(s) is (are)

(Assume $\Delta H^\circ$ and $\Delta S^\circ$ are independent of temperature and ratio of lnK at $T_1$ to lnK at $T_2$ is greater than $T_2/T_1$. Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.) **[JEE (Advanced) 2018]**

(A) $\Delta H^\circ < 0, \Delta S^\circ < 0$ (B) $\Delta G^\circ < 0, \Delta H^\circ > 0$ (C) $\Delta G^\circ < 0, \Delta S^\circ < 0$ (D) $\Delta G^\circ < 0, \Delta S^\circ > 0$

**CTS078**

**12.** Match the thermodynamic processes given under Column-I with the expressions given under Column-II.

| **Column - I** | **Column - II** |
|---|---|
| (1) Freezing of water at 273 K and 1 atm | (P) q = 0 |
| (2) Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions | (Q) w = 0 |
| (3) Mixing of equal volumes of two ideal gases at constant temperature and pressure in an isolated container | (R) $\Delta S_{sys} < 0$ |
| (4) Reversible heating of $H_2(g)$ at 1 atm from 300 K to 600 K followed by reversible cooling to 300 K at 1 atm | (S) $\Delta U = 0$ |
| | (T) $\Delta G = 0$ |

**[JEE (Advanced) 2015]**

(A) (1) → R, T ; (2) → P, Q, S ; (3) → P, Q, S ; (4) → P, Q, S, T

(B) (1) → P, T ; (2) → R, Q, S ; (3) → P, Q, S ; (4) → P, Q, S, T

(C) (1) → P, S ; (2) → R, Q, S ; (3) → P, Q, S, T ; (4) → P, Q, S

(D) (1) → Q, S ; (2) → T, Q, S ; (3) → P, Q, ; (4) → P, Q, S

**CTS072**

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# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | A | C | B | B | A | C | B | A | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | B | C | A | A | C | D | A | D | B |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| Ans. | C | D | B | C | A | D | B | C | C | |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Ans. | AD | BCD | ABD | A → Q; B → P, S; C → R, S | A → P, S; B → P, R, S; C → Q; D → R, S |

| Que. | 6 | 7 |
|---|---|---|
| Ans. | A → S; B → P; C → Q; D → R | B |

## EXERCISE (S)

**1.** 0 **2.** 5 **3.** 8 **4.** 436 **5.** 4

**6.** 4.00 **7.** (a) 2.10, (b) 1.50 **8.** 38.20 - 38.30

**9.** 10.32 - 10.33 **10.** (a) –250 J/K (b) 333.33 J/K (c) +83.33 J/K. Process is spontaneous.

**11.** –3.70 kJ **12.** 5.96 – 5.97 **13.** 2.10 **14.** 750.00

**15.** 27.00 **16.** 0.00, 1.00 **17.** –13.49

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## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 3 | 2 | 700 | 1 | 4 | 2850 | 2 | 1 | 2000 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 400 | 2 | 2 | 300 | 336 | –2.7 | 2 | 57 | 1411 | 2 |
| Que. | 21 | 22 | | | | | | | | |
| Ans. | 1 | 1 | | | | | | | | |

## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | ABD | C | A | 0.30 to 0.32 | 300 | AC | 8 | 10 | 935 |
| Que. | 11 | 12 | | | | | | | | |
| Ans. | AC | A | | | | | | | | |



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